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第二章　函数的导数(Derivative)

2.1　导数的定义(The Definition of Derivative)

Multiple Choice Questions

1. Which of the following values is the average rate of change of $f(x)=\sqrt{x+2}$ over the interval $[-1,2]$?

(A) -4　　(B) $-\frac{1}{3}$　　(C) $\frac{1}{3}$　　(D) 3

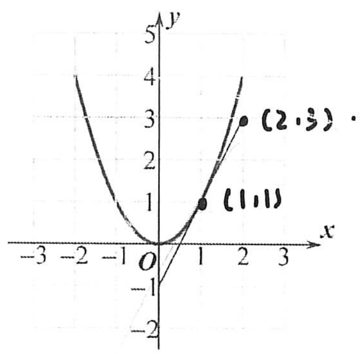
$f'(x)=\frac{f(x+h)-f(x)}{h}$

2. Let $f(x)=4-3x$. Which of the following is equal to $f'(0)$?

(A) 4　　(B) -4　　(C) -3　　(D) 4

$\frac{4-3x-3h-4+3x}{h}=-3$

3. The graph of f and its tangent line at $x=1$ is shown below, find $f'(1)=($ D $)$.



(A) 0　　(B) $\frac{1}{2}$　　(C) 1　　(D) 2

4. Let $f(x)=kx^3-2x,(k\neq 0)$. The average rate of $f(x)$ over the interval $[-2,-1]$ is 3, find k.

$3=-k+2$

$5=7k$

(A) $\frac{5}{7}$　　(B) $\frac{1}{7}$　　(C) $-\frac{1}{7}$　　(D) $-\frac{5}{7}$

5. **Caculator** Let $y=\tan e^x-x^2+1$. Find $\frac{dy}{dx}|_{x=0.5}=$

(A) -12.056　　(B) 0　　(C) 271.066　　(D) 273.066

6. **Caculator** The graph of $y=\sqrt{x^2+1}-3x$ crosses the x-axis at one point in the interval $[0,1]$. What is the slope of the graph at this point?

(A) 0.354　　(B) -0.001　　(C) -2.666　　(D) 0

7. The population of insects at time t days is modeled by the function P with first derivative $P'(t)=0.1t^2+10t+120$. Which of the following statements is true?

(A) At time $t=10$, the population of insects is 230.

(B) At time $t=10$, the population of insects is increasing at a rate of 230 insects per day.

(C) At time $t=10$, the population of insects is decreasing at a rate of 230 insects per day.

(D) At time $t=10$, the rate of change of the number of insects is increasing at a rate of 230 insects per day.

8. If the line tangent to the graph of a function g at the point (1,4) passes through the point(−2, −5), what is the slope of this line?

(A) 3 (B) −3 (C) $\frac{1}{3}$ (D) $-\frac{1}{3}$

Free Response Questions

9. Let $f(x)=x^2-2x$

(a) Find the average rate of change of f over [1,3].

(b) Find $\frac{f(2+h)-f(2)}{h}$.

(c) Find the instantaneous rate of change of f at $x=2$.

10. Use the definition of derivative to find the derivative of $f(x)=x^3$, then find $f'(2)$.